

Università degli Studi di Salerno

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CORSO DI LAUREA: INGEGNERIA INFORMATICA

# Software Engineering Software Testing

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# 1. Test Planning

Il gruppo in questa fase ha deciso di pianificare un'attività di testing estensiva che potesse andare a verificare ogni parte del codice, cosicché al cliente possa arrivare un prodotto finito e funzionante che non presenti alcun tipo di bug. Nella pratica, questo approccio si manifesta attraverso la completa verifica del modello. Nel contesto dell'applicazione della calcolatrice scientifica, il modello comprende la parte esecutiva e le strutture dati che rappresentano il nucleo dell'implementazione. Questi moduli sono quindi particolarmente propensi a incontrare problemi implementativi e sono soggetti a una verifica approfondita. Il gruppo ha, quindi, testato ogni metodo del modello in maniera esaustiva con diversi set di input. A differenza degli altri, la view e il controller verranno testati attraverso un'interazione con la GUI e senza l'utilizzo del codice in Java.

## 2. Test Cases Funzionali

I seguenti test cases funzionali mostrano diversi casi di interazione con la calcolatrice da parte dell'utente:

|                  |                                                                                               |                                                               |
|------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| FTC-01           | Operazioni aritmetiche                                                                        |                                                               |
| Precondizione    | Lo Stack è vuoto                                                                              |                                                               |
| Flusso di Eventi | Passi                                                                                         | Azioni                                                        |
|                  | 1                                                                                             | Viene inserita la cifra "8", il segno "+" e la cifra "3".     |
|                  | 2                                                                                             | Viene inserita la cifra "4", il segno "-" e la cifra "2".     |
|                  | 3                                                                                             | Si sceglie l'operazione sottrazione tramite il tasto "-".     |
|                  | 4                                                                                             | Viene mostrato a schermo "-4.0-5.0j".                         |
|                  | 5                                                                                             | Si sceglie l'operazione somma tramite il tasto "+".           |
|                  | 6                                                                                             | Viene mostrato a schermo "12.0+1.0j".                         |
|                  | 7                                                                                             | Viene inserita la cifra "2".                                  |
|                  | 8                                                                                             | Si sceglie l'operazione moltiplicazione tramite il tasto "*". |
|                  | 9                                                                                             | Viene mostrato a schermo "24.0+2.0j".                         |
|                  | 10                                                                                            | Si sceglie l'operazione "clear".                              |
|                  | 11                                                                                            | Viene inserita la cifra "0".                                  |
|                  | 12                                                                                            | Viene inserita la cifra "10".                                 |
|                  | 13                                                                                            | Si sceglie l'operazione divisione tramite il tasto "/".       |
|                  | 13                                                                                            | Viene mostrato a schermo "Divided by 0, ERROR".               |
| Postcondizione   | Visione della stringa "Stack Vuoto"                                                           |                                                               |
| Use Cases        | Inserimento operandi - Inserimento e gestione delle operazioni<br>- Manipolazione dello stack |                                                               |

|                  |                                                                                               |                                                                                 |
|------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| FTC-02           | Operazioni sullo stack                                                                        |                                                                                 |
| Precondizione    | Lo Stack è vuoto                                                                              |                                                                                 |
| Flusso di Eventi | Passi                                                                                         | Azioni                                                                          |
|                  | 1                                                                                             | Viene inserita la cifra "5", il segno "+" e la cifra "8".                       |
|                  | 2                                                                                             | Viene inserita la cifra "6", il segno "+" e la cifra "1".                       |
|                  | 3                                                                                             | Lo stack mostra nell'ordine, dall'alto verso il basso, "6.0+1.0j" e "5.0+8.0j". |
|                  | 3                                                                                             | Si sceglie l'operazione "swap".                                                 |
|                  | 4                                                                                             | Lo stack mostra "5.0+8.0j" e "6.0+1.0j".                                        |
|                  | 5                                                                                             | Si sceglie l'operazione "drop".                                                 |
|                  | 6                                                                                             | Lo stack mostra "6.0+1.0j".                                                     |
|                  | 7                                                                                             | Si sceglie l'operazione "swap".                                                 |
|                  | 8                                                                                             | Lo stack mostra "0.0+0.0j" e "6.0+1.0j".                                        |
|                  | 9                                                                                             | Si sceglie l'operazione "over".                                                 |
|                  | 10                                                                                            | Lo stack mostra "6.0+1.0j", "0.0+0.0j" e "6.0+1.0j".                            |
|                  | 11                                                                                            | Si sceglie l'operazione "clear".                                                |
|                  | 12                                                                                            | Lo stack è vuoto.                                                               |
|                  | 13                                                                                            | Si sceglie l'operazione "drop".                                                 |
|                  | 14                                                                                            | Viene mostrato a schermo "Errore, Stack Vuoto".                                 |
| Postcondizione   | Lo stack è vuoto                                                                              |                                                                                 |
| Use Cases        | Inserimento operandi - Inserimento e gestione delle operazioni<br>- Manipolazione dello stack |                                                                                 |

|                  |                             |                                                                                            |
|------------------|-----------------------------|--------------------------------------------------------------------------------------------|
| FTC-03           | Operazioni con le variabili |                                                                                            |
| Precondizione    | Lo stack è vuoto            |                                                                                            |
| Flusso di Eventi | Passi                       | Azioni                                                                                     |
|                  | 1                           | Si seleziona la variabile "x".                                                             |
|                  | 2                           | Si sceglie l'operazione ">var".                                                            |
|                  | 3                           | Viene mostrato a schermo "Errore, Stack Vuoto".                                            |
|                  | 4                           | Si preme il pulsante "AC".                                                                 |
|                  | 5                           | Viene inserita la cifra "3", il segno "+" e la cifra "2".                                  |
|                  | 5                           | Si seleziona la variabile "x".                                                             |
|                  | 6                           | Si sceglie l'operazione ">var".                                                            |
|                  | 7                           | Viene mostrato a schermo "Operazione di stack->x<br>Variabili Utilizzate: {x = 3.0+2.0j}". |
|                  | 8                           | Si seleziona la variabile "y".                                                             |
|                  | 9                           | Si sceglie l'operazione "<var".                                                            |
|                  | 10                          | Viene mostrato a schermo "Operazione di y->Stack<br>Variabili Utilizzate: {x= 3.0+2.0j}".  |
|                  | 11                          | Lo stack mostra "0.0+0.0j".                                                                |
|                  | 12                          | Si seleziona la variabile "x".                                                             |
|                  | 13                          | Si sceglie l'operazione "<var".                                                            |
|                  | 11                          | Lo stack mostra "3.0+2.0j", "0.0+0.0j".                                                    |
|                  | 12                          | Si seleziona la variabile "x".                                                             |
|                  | 13                          | Si sceglie l'operazione "+var".                                                            |
|                  | 14                          | Viene mostrato a schermo "Operazione di +x<br>Variabili Utilizzate: {x= 6.0+4.0j}".        |
|                  | 15                          | Lo stack mostra "0.0+0.0j".                                                                |

|                |                                                                                                     |                                                                                      |
|----------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|                | 16                                                                                                  | Si seleziona la variabile "x".                                                       |
|                | 17                                                                                                  | Si sceglie l'operazione "-var".                                                      |
|                | 18                                                                                                  | Viene mostrato a schermo "Operazione di -x<br>Variabili Utilizzate: {x= -6.0-4.0j}". |
|                | 19                                                                                                  | Lo stack mostra la stringa "Stack Vuoto".                                            |
|                | 20                                                                                                  | Si seleziona la variabile "a".                                                       |
|                | 21                                                                                                  | Si sceglie l'operazione "+var".                                                      |
|                | 22                                                                                                  | Viene mostrato a schermo "Variabile Vuota."                                          |
| Postcondizione | Lo stack è vuoto                                                                                    |                                                                                      |
| Use Cases      | Inserimento operandi - Manipolazione dello stack - Gestione delle variabili - Interfaccia variabili |                                                                                      |

### 3. Unit Test Cases

Negli unit test cases sono mostrati il funzionamento della singola unità di codice e il rispettivo risultato del test. Sono così definiti:

#### 3.1 Unit Test Cases: ComplexTest

|            |                                            |
|------------|--------------------------------------------|
| UTC-1.1    | ComplexTest.testGetRealPart                |
| Test items | classe ComplexTest, metodo testGetRealPart |
| Input      | Complex num = new Complex(23.5542, 53);    |
| Oracle     | double expResult = 23.5542;                |
| Report     | 100% PASSED                                |

|            |                                         |
|------------|-----------------------------------------|
| UTC-1.2    | ComplexTest.testGetJPart                |
| Test items | classe ComplexTest, metodo testGetJPart |
| Input      | Complex num = new Complex(23.5542, 0);  |
| Oracle     | double expResult = 0;                   |
| Report     | 100% PASSED                             |

|            |                                          |
|------------|------------------------------------------|
| UTC-1.3    | ComplexTest.testToString1                |
| Test items | classe ComplexTest, metodo testToString1 |
| Input      | Complex num = new Complex(532, 12);      |
| Oracle     | String expResult = "532.0+12.0j";        |
| Report     | 100% PASSED                              |

|            |                                          |
|------------|------------------------------------------|
| UTC-1.4    | ComplexTest.testToString2                |
| Test items | classe ComplexTest, metodo testToString2 |
| Input      | Complex num = new Complex(-32, -1);      |
| Oracle     | String expResult = "-32.0-1.0j";         |
| Report     | 100% PASSED                              |

|            |                                          |
|------------|------------------------------------------|
| UTC-1.5    | ComplexTest.testToString3                |
| Test items | classe ComplexTest, metodo testToString3 |
| Input      | Complex num = new Complex(432.124, 5.9); |
| Oracle     | String expResult = "432.124+5.9j";       |
| Report     | 100% PASSED                              |

|            |                                              |
|------------|----------------------------------------------|
| UTC-1.6    | ComplexTest.testToString4                    |
| Test items | classe ComplexTest, metodo testToString4     |
| Input      | Complex num = new Complex(-42.54, -12.2174); |
| Oracle     | String expResult = "-42.54-12.2174j";        |
| Report     | 100% PASSED                                  |

|            |                                          |
|------------|------------------------------------------|
| UTC-1.7    | ComplexTest.testToString5                |
| Test items | classe ComplexTest, metodo testToString5 |
| Input      | Complex num = new Complex(0, -12.634);   |
| Oracle     | String expResult = "0.0-12.634j";        |
| Report     | 100% PASSED                              |

|            |                                          |
|------------|------------------------------------------|
| UTC-1.8    | ComplexTest.testToString6                |
| Test items | classe ComplexTest, metodo testToString6 |
| Input      | Complex num = new Complex(0, 234);       |
| Oracle     | String expResult = "0.0+234.0j";         |
| Report     | 100% PASSED                              |

|            |                                          |
|------------|------------------------------------------|
| UTC-1.9    | ComplexTest.testToString7                |
| Test items | classe ComplexTest, metodo testToString7 |
| Input      | Complex num = new Complex(-312.5, 0);    |
| Oracle     | String expResult = "-312.5+0.0j";        |
| Report     | 100% PASSED                              |

|            |                                          |
|------------|------------------------------------------|
| UTC-1.10   | ComplexTest.testToString8                |
| Test items | classe ComplexTest, metodo testToString8 |
| Input      | Complex num = new Complex(75, 0);        |
| Oracle     | String expResult = "75.0+0.0j";          |
| Report     | 100% PASSED                              |

|            |                                          |
|------------|------------------------------------------|
| UTC-1.11   | ComplexTest.testToString9                |
| Test items | classe ComplexTest, metodo testToString9 |
| Input      | Complex num = new Complex(0, 0);         |
| Oracle     | String expResult = "0.0+0.0j";           |
| Report     | 100% PASSED                              |

|            |                                        |
|------------|----------------------------------------|
| UTC-1.12   | ComplexTest.testIsZero1                |
| Test items | classe ComplexTest, metodo testIsZero1 |
| Input      | Complex num = new Complex(532, 12);    |
| Oracle     | boolean expResult = false;             |
| Report     | 100% PASSED                            |

|            |                                        |
|------------|----------------------------------------|
| UTC-1.13   | ComplexTest.testIsZero2                |
| Test items | classe ComplexTest, metodo testIsZero2 |
| Input      | Complex num = new Complex(-32, -1);    |
| Oracle     | boolean expResult = false;             |
| Report     | 100% PASSED                            |

|            |                                        |
|------------|----------------------------------------|
| UTC-1.14   | ComplexTest.testIsZero3                |
| Test items | classe ComplexTest, metodo testIsZero3 |
| Input      | Complex num = new Complex(0, -12.634); |
| Oracle     | boolean expResult = false;             |

|            |                                        |
|------------|----------------------------------------|
| UTC-1.15   | ComplexTest.testIsZero4                |
| Test items | classe ComplexTest, metodo testIsZero4 |
| Input      | Complex num = new Complex(0, 234);     |
| Oracle     | boolean expResult = false;             |
| Report     | 100% PASSED                            |

|            |                                        |
|------------|----------------------------------------|
| UTC-1.16   | ComplexTest.testIsZero5                |
| Test items | classe ComplexTest, metodo testIsZero5 |
| Input      | Complex num = new Complex(-312.5, 0);  |
| Oracle     | boolean expResult = false;             |
| Report     | 100% PASSED                            |

|            |                                        |
|------------|----------------------------------------|
| UTC-1.17   | ComplexTest.testIsZero6                |
| Test items | classe ComplexTest, metodo testIsZero6 |
| Input      | Complex num = new Complex(75, 0);      |
| Oracle     | boolean expResult = false;             |
| Report     | 100% PASSED                            |

|            |                                        |
|------------|----------------------------------------|
| UTC-1.18   | ComplexTest.testIsZero7                |
| Test items | classe ComplexTest, metodo testIsZero7 |
| Input      | Complex num = new Complex(0, 0);       |
| Oracle     | boolean expResult = true;              |
| Report     | 100% PASSED                            |

|            |                                                                                    |
|------------|------------------------------------------------------------------------------------|
| UTC-1.19   | ComplexTest.testAdd1                                                               |
| Test items | classe ComplexTest, metodo testAdd1                                                |
| Input      | Complex num1 = new Complex(23.5, 53);<br>Complex num2 = new Complex(534.2, 23.98); |
| Oracle     | Complex expResult = new Complex(23.5+534.2, 53+23.98);                             |
| Report     | 100% PASSED                                                                        |

|            |                                                                                        |
|------------|----------------------------------------------------------------------------------------|
| UTC-1.20   | ComplexTest.testAdd2                                                                   |
| Test items | classe ComplexTest, metodo testAdd2                                                    |
| Input      | Complex num1 = new Complex(-235, -123);<br>Complex num2 = new Complex(-7.10, -1237.9); |
| Oracle     | Complex expResult = new Complex(-235-7.10, -123-1237.9);                               |
| Report     | 100% PASSED                                                                            |

|            |                                                                         |
|------------|-------------------------------------------------------------------------|
| UTC-1.21   | ComplexTest.testAdd3                                                    |
| Test items | classe ComplexTest, metodo testAdd3                                     |
| Input      | Complex num1 = new Complex(8, 0);<br>Complex num2 = new Complex(29, 0); |
| Oracle     | Complex expResult = new Complex(8+29, 0);                               |
| Report     | 100% PASSED                                                             |

|            |                                                                                |
|------------|--------------------------------------------------------------------------------|
| UTC-1.22   | ComplexTest.testAdd4                                                           |
| Test items | classe ComplexTest, metodo testAdd4                                            |
| Input      | Complex num1 = new Complex(0, 235);<br>Complex num2 = new Complex(0, -12.643); |
| Oracle     | Complex expResult = new Complex(0, 235-12.643);                                |
| Report     | 100% PASSED                                                                    |

|            |                                                                                 |
|------------|---------------------------------------------------------------------------------|
| UTC-1.23   | ComplexTest.testAdd5                                                            |
| Test items | classe ComplexTest, metodo testAdd5                                             |
| Input      | Complex num1 = new Complex(0, -235);<br>Complex num2 = new Complex(12.6, 43.1); |
| Oracle     | Complex expResult = new Complex(0+12.6, -235+43.1);                             |
| Report     | 100% PASSED                                                                     |

|            |                                                                                    |
|------------|------------------------------------------------------------------------------------|
| UTC-1.24   | ComplexTest.testSub1                                                               |
| Test items | classe ComplexTest, metodo testSub1                                                |
| Input      | Complex num1 = new Complex(23.5, 53);<br>Complex num2 = new Complex(534.2, 23.98); |

|        |                                                        |
|--------|--------------------------------------------------------|
| Oracle | Complex expResult = new Complex(23.5-534.2, 53-23.98); |
| Report | 100% PASSED                                            |

|            |                                                                                        |
|------------|----------------------------------------------------------------------------------------|
| UTC-1.25   | ComplexTest.testSub2                                                                   |
| Test items | classe ComplexTest, metodo testSub2                                                    |
| Input      | Complex num1 = new Complex(-235, -123);<br>Complex num2 = new Complex(-7.10, -1237.9); |
| Oracle     | Complex expResult = new Complex(-235+7.10, -123+1237.9);                               |
| Report     | 100% PASSED                                                                            |

|            |                                                                         |
|------------|-------------------------------------------------------------------------|
| UTC-1.26   | ComplexTest.testSub3                                                    |
| Test items | classe ComplexTest, metodo testSub3                                     |
| Input      | Complex num1 = new Complex(8, 0);<br>Complex num2 = new Complex(29, 0); |
| Oracle     | Complex expResult = new Complex(8-29, 0);                               |
| Report     | 100% PASSED                                                             |

|            |                                                                                |
|------------|--------------------------------------------------------------------------------|
| UTC-1.27   | ComplexTest.testSub4                                                           |
| Test items | classe ComplexTest, metodo testSub4                                            |
| Input      | Complex num1 = new Complex(0, 235);<br>Complex num2 = new Complex(0, -12.643); |
| Oracle     | Complex expResult = new Complex(0, 235+12.643);                                |
| Report     | 100% PASSED                                                                    |

|            |                                                                                 |
|------------|---------------------------------------------------------------------------------|
| UTC-1.28   | ComplexTest.testSub5                                                            |
| Test items | classe ComplexTest, metodo testSub5                                             |
| Input      | Complex num1 = new Complex(0, -235);<br>Complex num2 = new Complex(12.6, 43.1); |
| Oracle     | Complex expResult = new Complex(0-12.6, -235-43.1);                             |
| Report     | 100% PASSED                                                                     |

|            |                                                                                       |
|------------|---------------------------------------------------------------------------------------|
| UTC-1.29   | ComplexTest.testTimes1                                                                |
| Test items | classe ComplexTest, metodo testTimes1                                                 |
| Input      | Complex num1 = new Complex(23.5, 53);<br>Complex num2 = new Complex(534.2, 23.98);    |
| Oracle     | Complex expResult = new Complex((23.5*534.2) - (53*23.98),(23.5*23.98) + (53*534.2)); |
| Report     | 100% PASSED                                                                           |

|            |                                                                                                                          |
|------------|--------------------------------------------------------------------------------------------------------------------------|
| UTC-1.30   | ComplexTest.testTimes2                                                                                                   |
| Test items | classe ComplexTest, metodo testTimes2                                                                                    |
| Input      | Complex num1 = new Complex(-235, -123);<br>Complex num2 = new Complex(-7.10, -1237.9);                                   |
| Oracle     | Complex expResult = new Complex(((((-235) * (-7.10)) - ((-123) * (-1237.9))), ((-235) * (-1237.9))+((-123) * (-7.10)))); |
| Report     | 100% PASSED                                                                                                              |

|            |                                                                         |
|------------|-------------------------------------------------------------------------|
| UTC-1.31   | ComplexTest.testTimes3                                                  |
| Test items | classe ComplexTest, metodo testTimes3                                   |
| Input      | Complex num1 = new Complex(8, 0);<br>Complex num2 = new Complex(29, 0); |
| Oracle     | Complex expResult = new Complex(8*29, 0);                               |
| Report     | 100% PASSED                                                             |

|            |                                                                                |
|------------|--------------------------------------------------------------------------------|
| UTC-1.32   | ComplexTest.testTimes4                                                         |
| Test items | classe ComplexTest, metodo testTimes4                                          |
| Input      | Complex num1 = new Complex(0, 235);<br>Complex num2 = new Complex(0, -12.643); |
| Oracle     | Complex expResult = new Complex(-(235*(-12.643)), 0);                          |
| Report     | 100% PASSED                                                                    |

|            |                                       |
|------------|---------------------------------------|
| UTC-1.33   | ComplexTest.testTimes5                |
| Test items | classe ComplexTest, metodo testTimes5 |

|        |                                                                                 |
|--------|---------------------------------------------------------------------------------|
| Input  | Complex num1 = new Complex(0, -235);<br>Complex num2 = new Complex(12.6, 43.1); |
| Oracle | Complex expResult = new Complex(-(-235*43.1), (-235)*(12.6));                   |
| Report | 100% PASSED                                                                     |

|            |                                                                                    |
|------------|------------------------------------------------------------------------------------|
| UTC-1.34   | ComplexTest.testDivision1                                                          |
| Test items | classe ComplexTest, metodo testDivision1                                           |
| Input      | Complex num1 = new Complex(23.5, 53);<br>Complex num2 = new Complex(534.2, 23.98); |
| Oracle     | Complex expResult = new Complex(4.11294608, -8.25558051);                          |
| Report     | 100% PASSED                                                                        |

|            |                                                                                        |
|------------|----------------------------------------------------------------------------------------|
| UTC-1.35   | ComplexTest.testDivision2                                                              |
| Test items | classe ComplexTest, metodo testDivision2                                               |
| Input      | Complex num1 = new Complex(-235, -123);<br>Complex num2 = new Complex(-7.10, -1237.9); |
| Oracle     | Complex expResult = new Complex(2.18793914, 4.12248344);                               |
| Report     | 100% PASSED                                                                            |

|            |                                                                         |
|------------|-------------------------------------------------------------------------|
| UTC-1.36   | ComplexTest.testDivision3                                               |
| Test items | classe ComplexTest, metodo testDivision3                                |
| Input      | Complex num1 = new Complex(8, 0);<br>Complex num2 = new Complex(29, 0); |
| Oracle     | Complex expResult = new Complex(3.625, 0);                              |
| Report     | 100% PASSED                                                             |

|            |                                                                                |
|------------|--------------------------------------------------------------------------------|
| UTC-1.37   | ComplexTest.testDivision4                                                      |
| Test items | classe ComplexTest, metodo testDivision4                                       |
| Input      | Complex num1 = new Complex(0, 235);<br>Complex num2 = new Complex(0, -12.643); |
| Oracle     | Complex expResult = new Complex(-18.58736059, 0);                              |
| Report     | 100% PASSED                                                                    |

|            |                                                                                 |
|------------|---------------------------------------------------------------------------------|
| UTC-1.38   | ComplexTest.testDivision5                                                       |
| Test items | classe ComplexTest, metodo testDivision5                                        |
| Input      | Complex num1 = new Complex(0, -235);<br>Complex num2 = new Complex(12.6, 43.1); |
| Oracle     | Complex expResult = new Complex(-5.02313563, -1.46848049);                      |
| Report     | 100% PASSED                                                                     |

|            |                                                                      |
|------------|----------------------------------------------------------------------|
| UTC-1.39   | ComplexTest.testDivision6                                            |
| Test items | classe ComplexTest, metodo testDivision6                             |
| Input      | Complex num1= new Complex(4, 5);<br>Complex num2= new Complex(0, 0); |
| Oracle     | ArithmeticException                                                  |
| Report     | 100% PASSED                                                          |

|            |                                                                                                                     |
|------------|---------------------------------------------------------------------------------------------------------------------|
| UTC-1.40   | ComplexTest.testSqrt1                                                                                               |
| Test items | classe ComplexTest, metodo testSqrt1                                                                                |
| Input      | Complex num1 = new Complex(23.5, 53);                                                                               |
| Oracle     | Complex expResult = new Complex(Math.sqrt(magnitude) *<br>Math.cos(phase), Math.sqrt(magnitude) * Math.sin(phase)); |
| Report     | 100% PASSED                                                                                                         |

|            |                                                                                                                     |
|------------|---------------------------------------------------------------------------------------------------------------------|
| UTC-1.41   | ComplexTest.testSqrt2                                                                                               |
| Test items | classe ComplexTest, metodo testSqrt2                                                                                |
| Input      | Complex num1 = new Complex(-235, -123);                                                                             |
| Oracle     | Complex expResult = new Complex(Math.sqrt(magnitude) *<br>Math.cos(phase), Math.sqrt(magnitude) * Math.sin(phase)); |
| Report     | 100% PASSED                                                                                                         |

|            |                                                        |
|------------|--------------------------------------------------------|
| UTC-1.42   | ComplexTest.testSqrt3                                  |
| Test items | classe ComplexTest, metodo testSqrt3                   |
| Input      | Complex num1 = new Complex(8, 0);                      |
| Oracle     | Complex expResult = new Complex(Math.sqrt(magnitude) * |

|        |                                                                        |
|--------|------------------------------------------------------------------------|
|        | <code>Math.cos(phase), Math.sqrt(magnitude) * Math.sin(phase));</code> |
| Report | 100% PASSED                                                            |

|            |                                                                                                                                   |
|------------|-----------------------------------------------------------------------------------------------------------------------------------|
| UTC-1.43   | <b>ComplexTest.testSqrt4</b>                                                                                                      |
| Test items | classe ComplexTest, metodo testSqrt4                                                                                              |
| Input      | <code>Complex num1 = new Complex(0, 235);</code>                                                                                  |
| Oracle     | <code>Complex expResult = new Complex(Math.sqrt(magnitude) *<br/>Math.cos(phase), Math.sqrt(magnitude) * Math.sin(phase));</code> |
| Report     | 100% PASSED                                                                                                                       |

|            |                                                                                                                                   |
|------------|-----------------------------------------------------------------------------------------------------------------------------------|
| UTC-1.44   | <b>ComplexTest.testSqrt5</b>                                                                                                      |
| Test items | classe ComplexTest, metodo testSqrt5                                                                                              |
| Input      | <code>Complex num1 = new Complex(5, -235);</code>                                                                                 |
| Oracle     | <code>Complex expResult = new Complex(Math.sqrt(magnitude) *<br/>Math.cos(phase), Math.sqrt(magnitude) * Math.sin(phase));</code> |
| Report     | 100% PASSED+-                                                                                                                     |

|            |                                                                                 |
|------------|---------------------------------------------------------------------------------|
| UTC-1.45   | <b>ComplexTest.testReverse1</b>                                                 |
| Test items | classe ComplexTest, metodo testReverse1                                         |
| Input      | <code>Complex num1 = new Complex(23.5, 53);</code>                              |
| Oracle     | <code>double expResultRealPart = -23.5;<br/>double expResultJPart = -53;</code> |
| Report     | 100% PASSED+-                                                                   |

|            |                                                                               |
|------------|-------------------------------------------------------------------------------|
| UTC-1.46   | <b>ComplexTest.testReverse2</b>                                               |
| Test items | classe ComplexTest, metodo testReverse2                                       |
| Input      | <code>Complex num1 = new Complex(-235, -123);</code>                          |
| Oracle     | <code>double expResultRealPart = 235;<br/>double expResultJPart = 123;</code> |
| Report     | 100% PASSED+-                                                                 |

|            |                                                                |
|------------|----------------------------------------------------------------|
| UTC-1.47   | ComplexTest.testReverse3                                       |
| Test items | classe ComplexTest, metodo testReverse3                        |
| Input      | Complex num1 = new Complex(5, -235);                           |
| Oracle     | double expResultRealPart = -5;<br>double expResultJPart = 235; |
| Report     | 100% PASSED                                                    |

## 3.2 Unit Test Cases: OperationTest

|            |                                                                                                                                                                                              |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UTC-2.1    | OperationTest.testInsert                                                                                                                                                                     |
| Test items | classe OperationTest, metodo testInsert                                                                                                                                                      |
| Input      | Complex num1 = new Complex(2.0, -37.2);<br>operation.insert(num1);<br>Complex num2 = new Complex(-9.8,5);<br>operation.insert(num2);                                                         |
| Oracle     | double expResultRealPart1 = num1.getRealPart();<br>double expResultJPart1 = num1.getJPart();<br>double expResultRealPart2 = num2.getRealPart();<br>double expResultJPart2 = num2.getJPart(); |
| Report     | 100% PASSED                                                                                                                                                                                  |

|            |                                                                                                                                      |
|------------|--------------------------------------------------------------------------------------------------------------------------------------|
| UTC-2.2    | OperationTest.testAdd1                                                                                                               |
| Test items | classe OperationTest, metodo testAdd1                                                                                                |
| Input      | Complex num1 = new Complex(-5.9,46);<br>operation.insert(num1);<br>Complex num2 = new Complex(36.0,23.2);<br>operation.insert(num2); |
| Oracle     | Complex expResult = num1.add(num2);                                                                                                  |
| Report     | 100% PASSED                                                                                                                          |

|            |                                                                      |
|------------|----------------------------------------------------------------------|
| UTC-2.3    | OperationTest.testAdd2                                               |
| Test items | classe OperationTest, metodo testAdd2                                |
| Input      | Complex num1 = new Complex(-23.12,18.45);<br>operation.insert(num1); |
| Oracle     | Complex expResult = num1.add(num2);                                  |

|        |             |
|--------|-------------|
| Report | 100% PASSED |
|--------|-------------|

|            |                                       |
|------------|---------------------------------------|
| UTC-2.4    | OperationTest.testAdd3                |
| Test items | classe OperationTest, metodo testAdd3 |
| Input      | /                                     |
| Oracle     | EmptyStackException                   |
| Report     | 100% PASSED                           |

|            |                                                                                                                                          |
|------------|------------------------------------------------------------------------------------------------------------------------------------------|
| UTC-2.5    | OperationTest.testSub1                                                                                                                   |
| Test items | classe OperationTest, metodo testSub1                                                                                                    |
| Input      | Complex num1 = new Complex(8.0, -37.81);<br>operation.insert(num1);<br>Complex num2 = new Complex(101, -6.6);<br>operation.insert(num2); |
| Oracle     | Complex expResult = num2.sub(num1);                                                                                                      |
| Report     | 100% PASSED                                                                                                                              |

|            |                                                                       |
|------------|-----------------------------------------------------------------------|
| UTC-2.6    | OperationTest.testSub2                                                |
| Test items | classe OperationTest, metodo testSub2                                 |
| Input      | Complex num1 = new Complex(-1.2567, 25.9);<br>operation.insert(num1); |
| Oracle     | Complex expResult = num1.sub(num1);                                   |
| Report     | 100% PASSED                                                           |

|            |                                       |
|------------|---------------------------------------|
| UTC-2.7    | OperationTest.testSub3                |
| Test items | classe OperationTest, metodo testSub3 |
| Input      | /                                     |
| Oracle     | EmptyStackException                   |
| Report     | 100% PASSED                           |

|            |                                         |
|------------|-----------------------------------------|
| UTC-2.8    | OperationTest.testTimes1                |
| Test items | classe OperationTest, metodo testTimes1 |

|        |                                                                                |
|--------|--------------------------------------------------------------------------------|
| Input  | Complex num1 = new Complex(4.0, 4.0);<br>Complex num2 = new Complex(2.0, 2.0); |
| Oracle | Complex expResult = num1.times(num2);                                          |
| Report | 100% PASSED                                                                    |

|            |                                                                  |
|------------|------------------------------------------------------------------|
| UTC-2.9    | OperationTest.testTimes2                                         |
| Test items | classe OperationTest, metodo testTimes2                          |
| Input      | Complex num1 = new Complex(4.0, 4.0);<br>operation.insert(num1); |
| Oracle     | Complex expResult = num1.times(num1);                            |
| Report     | 100% PASSED                                                      |

|            |                                         |
|------------|-----------------------------------------|
| UTC-2.10   | OperationTest.testTimes3                |
| Test items | classe OperationTest, metodo testTimes3 |
| Input      | /                                       |
| Oracle     | EmptyStackException                     |
| Report     | 100% PASSED                             |

|            |                                                                                                                                         |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| UTC-2.11   | OperationTest.testDivision1                                                                                                             |
| Test items | classe OperationTest, metodo testDivision1                                                                                              |
| Input      | Complex num1 = new Complex(-3.0, -2.0);<br>operation.insert(num1);<br>Complex num2 = new Complex(4.0, -1.0);<br>operation.insert(num2); |
| Oracle     | Complex expResult = num1.division(num2);                                                                                                |
| Report     | 100% PASSED                                                                                                                             |

|            |                                                                  |
|------------|------------------------------------------------------------------|
| UTC-2.12   | OperationTest.testDivision2                                      |
| Test items | classe OperationTest, metodo testDivision2                       |
| Input      | Complex num1 = new Complex(5.0, 2.0);<br>operation.insert(num1); |
| Oracle     | Complex expResult = num1.division(num1);                         |
| Report     | 100% PASSED                                                      |

|            |                                            |
|------------|--------------------------------------------|
| UTC-2.13   | OperationTest.testDivision3                |
| Test items | classe OperationTest, metodo testDivision3 |
| Input      | /                                          |
| Oracle     | EmptyStackException                        |
| Report     | 100% PASSED                                |

|            |                                                                  |
|------------|------------------------------------------------------------------|
| UTC-2.14   | OperationTest.testSqrt1                                          |
| Test items | classe OperationTest, metodo testSqrt1                           |
| Input      | Complex num1 = new Complex(3.0, 4.0);<br>operation.insert(num1); |
| Oracle     | Complex expResult = num1.sqrt();                                 |
| Report     | 100% PASSED                                                      |

|            |                                        |
|------------|----------------------------------------|
| UTC-2.15   | OperationTest.testSqrt2                |
| Test items | classe OperationTest, metodo testSqrt2 |
| Input      | Complex num1 = new Complex(4.0, 4.0);  |
| Oracle     | EmptyStackException                    |
| Report     | 100% PASSED                            |

|            |                                                                  |
|------------|------------------------------------------------------------------|
| UTC-2.16   | OperationTest.testReverse1                                       |
| Test items | classe OperationTest, metodo testReverse1                        |
| Input      | Complex num1 = new Complex(3.0, 4.0);<br>operation.insert(num1); |
| Oracle     | Complex expResult = num1.reverse();                              |
| Report     | 100% PASSED                                                      |

|            |                                           |
|------------|-------------------------------------------|
| UTC-2.17   | OperationTest.testReverse2                |
| Test items | classe OperationTest, metodo testReverse2 |
| Input      | Complex num1 = new Complex(4.0, 4.0);     |
| Oracle     | EmptyStackException                       |

|        |             |
|--------|-------------|
| Report | 100% PASSED |
|--------|-------------|

### 3.3 Unit Test Cases: ParserTest

|            |                                          |
|------------|------------------------------------------|
| UTC-3.1    | ParserTest.testIsComplex1                |
| Test items | classe ParserTest, metodo testIsComplex1 |
| Input      | String s="3.4-4.5";                      |
| Oracle     | String expectedResultString=s;           |
| Report     | 100% PASSED                              |

|            |                                          |
|------------|------------------------------------------|
| UTC-3.2    | ParserTest.testIsComplex2                |
| Test items | classe ParserTest, metodo testIsComplex2 |
| Input      | String s="3.4";                          |
| Oracle     | String expectedResultString=s + "+0";    |
| Report     | 100% PASSED                              |

|            |                                           |
|------------|-------------------------------------------|
| UTC-3.3    | ParserTest.testIsRealPart1                |
| Test items | classe ParserTest, metodo testIsRealPart1 |
| Input      | String s = "-24.87+15.01";                |
| Oracle     | String expectedResult = "-24.87";         |
| Report     | 100% PASSED                               |

|            |                                           |
|------------|-------------------------------------------|
| UTC-3.4    | ParserTest.testIsRealPart2                |
| Test items | classe ParserTest, metodo testIsRealPart2 |
| Input      | String s = "4.8-15.81";                   |
| Oracle     | String expectedResult = "4.8";            |
| Report     | 100% PASSED                               |

|            |                                        |
|------------|----------------------------------------|
| UTC-3.5    | ParserTest.testIsJPart1                |
| Test items | classe ParserTest, metodo testIsJPart1 |

|        |                          |
|--------|--------------------------|
| Input  | String s = "-4+3";       |
| Oracle | String expResult = "+3"; |
| Report | 100% PASSED              |

|            |                                        |
|------------|----------------------------------------|
| UTC-3.6    | ParserTest.testIsJPart2                |
| Test items | classe ParserTest, metodo testIsJPart2 |
| Input      | String s = "+14-23";                   |
| Oracle     | String expResult = "-23";              |
| Report     | 100% PASSED                            |

|            |                                                                               |
|------------|-------------------------------------------------------------------------------|
| UTC-3.7    | ParserTest.testParse1                                                         |
| Test items | classe ParserTest, metodo testParse1                                          |
| Input      | String s="34";<br>String svolgi="numero";<br>Complex num= new Complex(34, 0); |
| Oracle     | String expStringResult= num.toString();                                       |
| Report     | 100% PASSED                                                                   |

|            |                                                                                       |
|------------|---------------------------------------------------------------------------------------|
| UTC-3.8    | ParserTest.testParse2                                                                 |
| Test items | classe ParserTest, metodo testParse2                                                  |
| Input      | String svolgi="+";<br>Complex num= new Complex(34.56, -67.9);<br>stack.addFirst(num); |
| Oracle     | Complex expResult= num.add(num);                                                      |
| Report     | 100% PASSED                                                                           |

|            |                                                                                       |
|------------|---------------------------------------------------------------------------------------|
| UTC-3.9    | ParserTest.testParse3                                                                 |
| Test items | classe ParserTest, metodo testParse3                                                  |
| Input      | String svolgi="-";<br>Complex num= new Complex(34.56, -67.9);<br>stack.addFirst(num); |
| Oracle     | Complex expResult= num.sub(num);                                                      |
| Report     | 100% PASSED                                                                           |

|            |                                                                                       |
|------------|---------------------------------------------------------------------------------------|
| UTC-3.10   | ParserTest.testParse4                                                                 |
| Test items | classe ParserTest, metodo testParse4                                                  |
| Input      | String svolgi="*";<br>Complex num= new Complex(34.56, -67.9);<br>stack.addFirst(num); |
| Oracle     | Complex expResult= num.times(num);                                                    |
| Report     | 100% PASSED                                                                           |

|            |                                                                                       |
|------------|---------------------------------------------------------------------------------------|
| UTC-3.11   | ParserTest.testParse5                                                                 |
| Test items | classe ParserTest, metodo testParse5                                                  |
| Input      | String svolgi="/";<br>Complex num= new Complex(34.56, -67.9);<br>stack.addFirst(num); |
| Oracle     | Complex expResult= num.division(num);                                                 |
| Report     | 100% PASSED                                                                           |

|            |                                                                                          |
|------------|------------------------------------------------------------------------------------------|
| UTC-3.12   | ParserTest.testParse6                                                                    |
| Test items | classe ParserTest, metodo testParse6                                                     |
| Input      | String svolgi="sqrt";<br>Complex num= new Complex(34.56, -67.9);<br>stack.addFirst(num); |
| Oracle     | Complex expResult= num.sqrt();                                                           |
| Report     | 100% PASSED                                                                              |

|            |                                                                                        |
|------------|----------------------------------------------------------------------------------------|
| UTC-3.13   | ParserTest.testParse7                                                                  |
| Test items | classe ParserTest, metodo testParse7                                                   |
| Input      | String svolgi="+-";<br>Complex num= new Complex(34.56, -67.9);<br>stack.addFirst(num); |
| Oracle     | Complex expResult= num.reverse();                                                      |
| Report     | 100% PASSED                                                                            |

|            |                                                                                                                                                              |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UTC-3.14   | ParserTest.testParse8                                                                                                                                        |
| Test items | classe ParserTest, metodo testParse8                                                                                                                         |
| Input      | String svolgi="drop";<br>Complex num= new Complex(34.56, -67.9);<br>stack.addFirst(num1);<br>Complex num2= new Complex(46.9, 90.1);<br>stack.addFirst(num2); |
| Oracle     | double expResultRealPart = num1.getRealPart();<br>double expResultJPart = num1.getJPart();                                                                   |
| Report     | 100% PASSED                                                                                                                                                  |

|            |                                                                                            |
|------------|--------------------------------------------------------------------------------------------|
| UTC-3.15   | ParserTest.testParse9                                                                      |
| Test items | classe ParserTest, metodo testParse9                                                       |
| Input      | String svolgi="dup";<br>Complex num1= new Complex(34.56, -67.9);<br>stack.addFirst(num1);  |
| Oracle     | double expResultRealPart = num1.getRealPart();<br>double expResultJPart = num1.getJPart(); |
| Report     | 100% PASSED                                                                                |

|            |                                                                                                                                                               |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UTC-3.16   | ParserTest.testParse10                                                                                                                                        |
| Test items | classe ParserTest, metodo testParse10                                                                                                                         |
| Input      | String svolgi="swap";<br>Complex num1= new Complex(34.56, -67.9);<br>stack.addFirst(num1);<br>Complex num2= new Complex(46.9, 90.1);<br>stack.addFirst(num2); |
| Oracle     | double expResultRealPart = num1.getRealPart();<br>double expResultJPart = num1.getJPart();                                                                    |
| Report     | 100% PASSED                                                                                                                                                   |

|            |                                                                                                                                       |
|------------|---------------------------------------------------------------------------------------------------------------------------------------|
| UTC-3.17   | ParserTest.testParse11                                                                                                                |
| Test items | classe ParserTest, metodo testParse11                                                                                                 |
| Input      | String svolgi="clear";<br>Complex num1= new Complex(34.56, -67.9);<br>stack.addFirst(num1);<br>Complex num2= new Complex(46.9, 90.1); |

|        |                           |
|--------|---------------------------|
|        | stack.addFirst(num2);     |
| Oracle | Complex num = new (0, 0); |
| Report | 100% PASSED               |

|            |                                                                                                                                                               |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UTC-3.18   | ParserTest.testParse12                                                                                                                                        |
| Test items | classe ParserTest, metodo testParse12                                                                                                                         |
| Input      | String svolgi="over";<br>Complex num1= new Complex(34.56, -67.9);<br>stack.addFirst(num1);<br>Complex num2= new Complex(46.9, 90.1);<br>stack.addFirst(num2); |
| Oracle     | double expResultRealPart = num1.getRealPart();<br>double expResultJPart = num1.getJPart();                                                                    |
| Report     | 100% PASSED                                                                                                                                                   |

|            |                                                                                                                       |
|------------|-----------------------------------------------------------------------------------------------------------------------|
| UTC-3.19   | ParserTest.testParse13                                                                                                |
| Test items | classe ParserTest, metodo testParse13                                                                                 |
| Input      | String svolgi="stackToVar";<br>String input="a";<br>Complex num1= new Complex(34.56, -67.9);<br>stack.addFirst(num1); |
| Oracle     | Complex num = new (0, 0);                                                                                             |
| Report     | 100% PASSED                                                                                                           |

|            |                                                                                                                       |
|------------|-----------------------------------------------------------------------------------------------------------------------|
| UTC-3.20   | ParserTest.testParse14                                                                                                |
| Test items | classe ParserTest, metodo testParse14                                                                                 |
| Input      | String svolgi="varToStack";<br>String input="a";<br>Complex num1= new Complex(34.56, -67.9);<br>stack.addFirst(num1); |
| Oracle     | Complex num = new (1, 0);                                                                                             |
| Report     | 100% PASSED                                                                                                           |

|            |                                       |
|------------|---------------------------------------|
| UTC-3.21   | ParserTest.testParse15                |
| Test items | classe ParserTest, metodo testParse15 |

|        |                                                                                                                 |
|--------|-----------------------------------------------------------------------------------------------------------------|
| Input  | String input="a";<br>String svolgi="addVar";<br>Complex num2= new Complex(8.2, 38.45);<br>stack.addFirst(num2); |
| Oracle | VarException                                                                                                    |
| Report | 100% PASSED                                                                                                     |

|            |                                                                                                                 |
|------------|-----------------------------------------------------------------------------------------------------------------|
| UTC-3.22   | ParserTest.testParse16                                                                                          |
| Test items | classe ParserTest, metodo testParse16                                                                           |
| Input      | String input="a";<br>String svolgi="addVar";<br>Complex num2= new Complex(8.2, 38.45);<br>stack.addFirst(num2); |
| Oracle     | VarException                                                                                                    |
| Report     | 100% PASSED                                                                                                     |

|            |                                              |
|------------|----------------------------------------------|
| UTC-3.23   | ParserTest.testCheckerNumber2                |
| Test items | classe ParserTest, metodo testCheckerNumber2 |
| Input      | /                                            |
| Oracle     | InputException                               |
| Report     | 100% PASSED                                  |

|            |                                              |
|------------|----------------------------------------------|
| UTC-3.24   | ParserTest.testCheckerNumber3                |
| Test items | classe ParserTest, metodo testCheckerNumber3 |
| Input      | /                                            |
| Oracle     | InputException                               |
| Report     | 100% PASSED                                  |

|            |                                              |
|------------|----------------------------------------------|
| UTC-3.25   | ParserTest.testCheckerNumber4                |
| Test items | classe ParserTest, metodo testCheckerNumber4 |
| Input      | /                                            |
| Oracle     | OverflowException                            |
| Report     | 100% PASSED                                  |

|            |                                           |
|------------|-------------------------------------------|
| UTC-3.26   | ParserTest.testCheckerVar1                |
| Test items | classe ParserTest, metodo testCheckerVar1 |
| Input      | /                                         |
| Oracle     | InputException                            |
| Report     | 100% PASSED                               |

|            |                                           |
|------------|-------------------------------------------|
| UTC-3.27   | ParserTest.testCheckerVar2                |
| Test items | classe ParserTest, metodo testCheckerVar2 |
| Input      | /                                         |
| Oracle     | InputException                            |
| Report     | 100% PASSED                               |

|            |                                           |
|------------|-------------------------------------------|
| UTC-3.28   | ParserTest.testCheckerVar3                |
| Test items | classe ParserTest, metodo testCheckerVar1 |
| Input      | /                                         |
| Oracle     | InputException                            |
| Report     | 100% PASSED                               |

|            |                                           |
|------------|-------------------------------------------|
| UTC-3.29   | ParserTest.testCheckerVar4                |
| Test items | classe ParserTest, metodo testCheckerVar1 |
| Input      | /                                         |
| Oracle     | InputException                            |
| Report     | 100% PASSED                               |

|            |                                                                   |
|------------|-------------------------------------------------------------------|
| UTC-3.30   | ParserTest.testVarMapToString                                     |
| Test items | classe ParserTest, metodo testVarMapToString                      |
| Input      | Complex num1= new Complex(34.56, -67.9);<br>stack.addFirst(num1); |
| Oracle     | String expStringResult="\nVariabili Utilizzate: \n{a="+ num1+"}"; |
| Report     | 100% PASSED                                                       |

### 3.4 Unit Test Cases: StackTest

|            |                                                                                                                                     |
|------------|-------------------------------------------------------------------------------------------------------------------------------------|
| UTC-4.1    | StackTest.testClear1                                                                                                                |
| Test items | classe StackTest, metodo testClear1                                                                                                 |
| Input      | Complex num1 = new Complex(-3.0, -2.0);<br>stack.addFirst(num1);<br>Complex num2 = new Complex(4.0, -1.0);<br>stack.addFirst(num2); |
| Oracle     | Stack vuoto                                                                                                                         |
| Report     | 100% PASSED                                                                                                                         |

|            |                                     |
|------------|-------------------------------------|
| UTC-4.2    | StackTest.testClear2                |
| Test items | classe StackTest, metodo testClear2 |
| Input      | /                                   |
| Oracle     | EmptyStackExcpetion                 |
| Report     | 100% PASSED                         |

|            |                                                                                                                                     |
|------------|-------------------------------------------------------------------------------------------------------------------------------------|
| UTC-4.3    | StackTest.testDrop1                                                                                                                 |
| Test items | classe StackTest, metodo testDrop1                                                                                                  |
| Input      | Complex num1 = new Complex(-3.0, -2.0);<br>stack.addFirst(num1);<br>Complex num2 = new Complex(4.0, -1.0);<br>stack.addFirst(num2); |
| Oracle     | double expResultRealPart = num1.getRealPart();<br>double expResultJPart = num1.getJPart();                                          |
| Report     | 100% PASSED                                                                                                                         |

|            |                                    |
|------------|------------------------------------|
| UTC-4.4    | StackTest.testDrop2                |
| Test items | classe StackTest, metodo testDrop2 |
| Input      | /                                  |
| Oracle     | EmptyStackExcpetion                |
| Report     | 100% PASSED                        |

|            |                                                                                            |
|------------|--------------------------------------------------------------------------------------------|
| UTC-4.5    | StackTest.testDup1                                                                         |
| Test items | classe StackTest, metodo testDup1                                                          |
| Input      | Complex num1 = new Complex(-15.8, 23.02);<br>stack.addFirst(num1);                         |
| Oracle     | double expResultRealPart = num1.getRealPart();<br>double expResultJPart = num1.getJPart(); |
| Report     | 100% PASSED                                                                                |

|            |                                                                        |
|------------|------------------------------------------------------------------------|
| UTC-4.6    | StackTest.testDup2                                                     |
| Test items | classe StackTest, metodo testDup1                                      |
| Input      | /                                                                      |
| Oracle     | Complex num1 = new Complex(0, 0);<br>Complex num2 = new Complex(0, 0); |
| Report     | 100% PASSED                                                            |

|            |                                                                                                                                                                                              |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UTC-4.7    | StackTest.testSwap1                                                                                                                                                                          |
| Test items | classe StackTest, metodo testSwap1                                                                                                                                                           |
| Input      | Complex num1 = new Complex(-3.0, -2.0);<br>stack.addFirst(num1);<br>Complex num2 = new Complex(4.0, -1.0);<br>stack.addFirst(num2);                                                          |
| Oracle     | double expResultRealPart1 = num1.getRealPart();<br>double expResultJPart1 = num1.getJPart();<br>double expResultRealPart2 = num2.getRealPart();<br>double expResultJPart2 = num2.getJPart(); |
| Report     | 100% PASSED                                                                                                                                                                                  |

|            |                                                                                            |
|------------|--------------------------------------------------------------------------------------------|
| UTC-4.8    | StackTest.testSwap2                                                                        |
| Test items | classe StackTest, metodo testSwap2                                                         |
| Input      | Complex num1 = new Complex(5.78, 89.33);<br>stack.addFirst(num1);                          |
| Oracle     | double expResultRealPart = num1.getRealPart();<br>double expResultJPart = num1.getJPart(); |
| Report     | 100% PASSED                                                                                |

|            |                                                             |
|------------|-------------------------------------------------------------|
| UTC-4.9    | StackTest.testSwap3                                         |
| Test items | classe StackTest, metodo testIsSwap3                        |
| Input      | /                                                           |
| Oracle     | double expResultRealPart = 0;<br>double expResultJPart = 0; |
| Report     | 100% PASSED                                                 |

|            |                                                                                                                                                                                              |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UTC-4.10   | StackTest.testOver1                                                                                                                                                                          |
| Test items | classe StackTest, metodo testIsOver1                                                                                                                                                         |
| Input      | Complex num1 = new Complex(-3.0, -2.0);<br>stack.addFirst(num1);<br>Complex num2 = new Complex(4.0, -1.0);<br>stack.addFirst(num2);                                                          |
| Oracle     | double expResultRealPart1 = num1.getRealPart();<br>double expResultJPart1 = num1.getJPart();<br>double expResultRealPart2 = num2.getRealPart();<br>double expResultJPart2 = num2.getJPart(); |
| Report     | 100% PASSED                                                                                                                                                                                  |

|            |                                                                                            |
|------------|--------------------------------------------------------------------------------------------|
| UTC-4.11   | StackTest.testOver2                                                                        |
| Test items | classe StackTest, metodo testIsOver2                                                       |
| Input      | Complex num1 = new Complex(5.78, 89.33);<br>stack.addFirst(num1);                          |
| Oracle     | double expResultRealPart = num1.getRealPart();<br>double expResultJPart = num1.getJPart(); |
| Report     | 100% PASSED                                                                                |

|            |                                                             |
|------------|-------------------------------------------------------------|
| UTC-4.12   | StackTest.testOver3                                         |
| Test items | classe StackTest, metodo testIsOver3                        |
| Input      | /                                                           |
| Oracle     | double expResultRealPart = 0;<br>double expResultJPart = 0; |
| Report     | 100% PASSED                                                 |

### 3.5 Unit Test Cases: VarMapTest

|            |                                                                                          |
|------------|------------------------------------------------------------------------------------------|
| UTC-5.1    | VarMapTest.testStackToVar1                                                               |
| Test items | classe VarMapTest, metodo testStackToVar1                                                |
| Input      | Complex num = new Complex(-48.28, -62.000);                                              |
| Oracle     | double expResultRealPart = num.getRealPart();<br>double expResultJPart = num.getJPart(); |
| Report     | 100% PASSED                                                                              |

|            |                                           |
|------------|-------------------------------------------|
| UTC-5.2    | VarMapTest.testStackToVar2                |
| Test items | classe VarMapTest, metodo testStackToVar2 |
| Input      | /                                         |
| Oracle     | EmptyStackException                       |
| Report     | 100% PASSED                               |

|            |                                                                                          |
|------------|------------------------------------------------------------------------------------------|
| UTC-5.3    | VarMapTest.testVarToStack1                                                               |
| Test items | classe VarMapTest, metodo testVarToStack1                                                |
| Input      | Complex num = new Complex(48.28, 62.000);                                                |
| Oracle     | double expResultRealPart = num.getRealPart();<br>double expResultJPart = num.getJPart(); |
| Report     | 100% PASSED                                                                              |

|            |                                           |
|------------|-------------------------------------------|
| UTC-5.4    | VarMapTest.testVarToStack2                |
| Test items | classe VarMapTest, metodo testVarToStack2 |
| Input      | Complex zero = new Complex(0, 0);         |
| Oracle     | Complex num = new Complex(0, 0);          |
| Report     | 100% PASSED                               |

|            |                                           |
|------------|-------------------------------------------|
| UTC-5.5    | VarMapTest.testAddVar1                    |
| Test items | classe VarMapTest, metodo testAddVar1     |
| Input      | Complex num1 = new Complex(-3.34, 24.67); |

|        |                                                                                                                         |
|--------|-------------------------------------------------------------------------------------------------------------------------|
|        | stack.addFirst(num1);<br>Complex num2 = new Complex(44.0, -1.0);                                                        |
| Oracle | double expResultRealPart = result.getRealPart();<br>double expResultJPart = result.getJPart();<br>stack.addFirst(num2); |
| Report | 100% PASSED                                                                                                             |

|            |                                                                                                |
|------------|------------------------------------------------------------------------------------------------|
| UTC-5.6    | VarMapTest.testAddVar2                                                                         |
| Test items | classe VarMapTest, metodo testAddVar2                                                          |
| Input      | Complex num1 = new Complex(-3.34, 24.67);<br>stack.addFirst(num1);                             |
| Oracle     | double expResultRealPart = result.getRealPart();<br>double expResultJPart = result.getJPart(); |
| Report     | 100% PASSED                                                                                    |

|            |                                                                    |
|------------|--------------------------------------------------------------------|
| UTC-5.7    | VarMapTest.testAddVar3                                             |
| Test items | classe VarMapTest, metodo testAddVar3                              |
| Input      | Complex num1 = new Complex(-3.34, 24.67);<br>stack.addFirst(num1); |
| Oracle     | VarException                                                       |
| Report     | 100% PASSED                                                        |

|            |                                                                                                                                        |
|------------|----------------------------------------------------------------------------------------------------------------------------------------|
| UTC-5.8    | VarMapTest.testSubVar1                                                                                                                 |
| Test items | classe VarMapTest, metodo testSubVar1                                                                                                  |
| Input      | Complex num1 = new Complex(-3.34, 24.67);<br>stack.addFirst(num1);<br>Complex num2 = new Complex(44.0, -1.0);<br>stack.addFirst(num2); |
| Oracle     | double expResultRealPart = result.getRealPart();<br>double expResultJPart = result.getJPart();                                         |
| Report     | 100% PASSED                                                                                                                            |

|            |                                       |
|------------|---------------------------------------|
| UTC-5.9    | VarMapTest.testSubVar2                |
| Test items | classe VarMapTest, metodo testSubVar2 |

|        |                                                                                                |
|--------|------------------------------------------------------------------------------------------------|
| Input  | Complex num1 = new Complex(-3.34, 24.67);                                                      |
| Oracle | double expResultRealPart = result.getRealPart();<br>double expResultJPart = result.getJPart(); |
| Report | 100% PASSED                                                                                    |

|            |                                                                    |
|------------|--------------------------------------------------------------------|
| UTC-5.10   | VarMapTest.testSubVar3                                             |
| Test items | classe VarMapTest, metodo testSubVar3                              |
| Input      | Complex num1 = new Complex(-3.34, 24.67);<br>stack.addFirst(num1); |
| Oracle     | VarException                                                       |
| Report     | 100% PASSED                                                        |

|            |                                                                                      |
|------------|--------------------------------------------------------------------------------------|
| UTC-5.11   | VarMapTest.testToString                                                              |
| Test items | classe VarMapTest, metodo testToString                                               |
| Input      | Complex num1 = new Complex(-3.34, 24.67);<br>Complex num2 = new Complex(44.0, -1.0); |
| Oracle     | String expResultString="\nVariabili Utilizzate: \n{a="+ num1+ ", c=" +<br>num2 +"}"; |
| Report     | 100% PASSED                                                                          |

## 4. Aggiornamento matrice della tracciabilità

### Legenda:



: Requisito non ancora soddisfatto nella fase corrispondente.



: Requisito soddisfatto nella fase corrispondente.

**ID** : Identificativo del paragrafo dove viene soddisfatto il Requisito.

| Requisito | Design                                                                                            | Implementazione                                                                               | Testing                                                                                              | Requisiti Correlati |
|-----------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------|
| 1         | <br>ID: 2. e 3.1 | <br>ID: 3.   | <br>ID: 3.1         | ...                 |
| 2         | <br>ID: 2. e 3.2 | <br>ID: 3.   | <br>ID: 3.2         | 1                   |
| 3         | <br>ID: 2.       | <br>ID: 3.   | <br>ID: 3.2         | 1                   |
| 4         | <br>ID: 2.     | <br>ID: 3. | <br>ID: 3.1 e 3.2 | 1, 2, 3             |
| 5         | <br>ID: 2.     | <br>ID: 3. | <br>ID: 3.1 e 3.2 | 1, 2, 3             |
| 6         | <br>ID: 2.     | <br>ID: 3. | <br>ID: 3.1 e 3.2 | 1, 2, 3             |
| 7         | <br>ID: 2.     | <br>ID: 3. | <br>ID: 3.1 e 3.2 | 1, 2, 3             |
| 8         | <br>ID: 2.     | <br>ID: 3. | <br>ID: 3.1 e 3.2 | 1, 2, 3             |
| 9         | <br>ID: 2.     | <br>ID: 3. | <br>ID: 3.1 e 3.2 | 1, 2, 3             |

|    |                                                                                                     |                                                                                               |                                                                                                     |                      |
|----|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------|
| 10 | <br>ID: 2. e 3.3   | <br>ID: 3.   | <br>ID: 3.4       | 1, 2, 3              |
| 11 | <br>ID: 4.1        | <br>ID: 2.   | <br>ID: 3.3       | 1-10                 |
| 12 | <br>ID: 4.1        | <br>ID: 2.   | <br>ID: 3.3       | 1, 2, 11             |
| 13 | <br>ID: 2.         | <br>ID: 3.   | <br>ID: 3.4       | 10, 11               |
| 14 | <br>ID: 2.         | <br>ID: 3.   | <br>ID: 3.4       | 10, 11               |
| 15 | <br>ID: 2.         | <br>ID: 3.   | <br>ID: 3.4       | 10, 11               |
| 16 | <br>ID: 2.         | <br>ID: 3.   | <br>ID: 3.4       | 10, 11               |
| 17 | <br>ID: 2.       | <br>ID: 3. | <br>ID: 3.4     | 10, 11               |
| 18 | <br>ID: 2. e 3.4 | <br>ID: 3. | <br>ID: 3.3     | 1, 2, 10, 11         |
| 19 | <br>ID: 4.1 e 3. | <br>ID: 2. | <br>ID: 3.3     | 12, 18               |
| 20 | <br>ID: 2.       | <br>ID: 3. | <br>ID: 3.5     | 1, 2, 10, 11, 18, 19 |
| 21 | <br>ID: 2.       | <br>ID: 3. | <br>ID: 3.5     | 1, 2, 10, 11, 18, 19 |
| 22 | <br>ID: 2.       | <br>ID: 3. | <br>ID: 3.5     | 1, 2, 10, 11, 18, 19 |
| 23 | <br>ID: 2.       | <br>ID: 3. | <br>ID: 3.5     | 1, 2, 10, 11, 18, 19 |
| 24 | <br>ID: 2.       | <br>ID: 3. | <br>ID: 2. e 3. | 1, 2, 10, 18, 19     |

|    |                                                                                                           |                                                                                                         |                                                                                    |                          |
|----|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------|
| 25 | <br><b>ID: 4.1 e 4.2</b> | <br><b>ID: 4.</b>      |  | 1, 2, 10, 11, 12, 18, 19 |
| 26 | <br><b>ID: 2. e 3.</b>   | <br><b>ID: 1. e 3.</b> |  | ...                      |
| 27 | <br><b>ID: 2.</b>        | <br><b>ID: 1. e 3.</b> |  | Tutti                    |